PFAs (Per- and Polyfluoroalkyl Substances)

Author: Usense Healthcare

Identity Sheet

Property	Value
Name	Per- and Polyfluoroalkyl Substances (PFAS)
Other names	Forever chemicals, PFCs, PFOS, PFOA, GenX
Chemical formula	Variable; general: CnF(2n+1)-R where R = functional group
Key species	PFOS ($C_8HF_{17}O_3S$, MW 500.13); PFOA ($C_8HF_{15}O_2$, MW 414.07)
CAS numbers	PFOS: 1763-23-1; PFOA: 335-67-1
Nature	Class of >12,000 synthetic fluorinated organic compounds
Solubility	Variable (amphiphilic); PFOA ~3.4 g/L; PFOS ~0.57 g/L

General structural formula:

```
F F F F F F F F
| | | | | | | |
C-C-C-C-C-C-C-C-SO3(-) (PFOS: perfluorooctane sulfonate)
| | | | | | | |
F F F F F F F F
```

PFAS are a large class of synthetic organofluorine compounds characterised by strong C-F bonds that make them extremely persistent in the environment ("forever chemicals"). They are used in non-stick coatings (Teflon), firefighting foams (AFFF), waterproof textiles, food packaging, and industrial processes. Due to their extreme persistence, PFAS are ubiquitous environmental contaminants detected in the blood and urine of virtually the entire global population. Urinary PFAS are emerging biomonitoring targets.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
PFOS	$C_8F_{17}SO_3^-$	499.12	Perfluorooctane sulfonate (long-chain, legacy; half-life ~5 years in humans)
PFOA	$C_7F_{15}COOH$	414.07	Perfluorooctanoic acid (long-chain, legacy; half-life ~3.5 years)
GenX (HFPO-DA)	$C_6HF_{11}O_3$	330.05	Short-chain replacement; shorter half-life but still persistent

Molecule	Formula	MW (g/mol)	Key Difference
PFBS	$C_4F_9SO_3^-$	299.09	Short-chain sulfonate; faster excretion
Fluoride (F^-)	F^-	19.00	Simple inorganic ion; completely different from organofluorine PFAS

Medical Information

Origin

Endogenous PFAS are **not endogenous**. They are entirely anthropogenic synthetic chemicals.

Exogenous

- **Drinking water:** Contaminated by industrial discharge, AFFF firefighting foam runoff. EPA MCL for PFOA and PFOS: 4 ng/L (2024 rule).
- **Food:** Migration from food packaging; bioaccumulation in fish, meat, dairy.
- **Consumer products:** Non-stick cookware, stain-resistant fabrics, cosmetics.
- **Occupational:** Manufacturing workers, firefighters (AFFF exposure).
- **Air/dust:** Indoor dust from treated textiles and carpets.

Primary & Secondary Biological Roles

PFAS have **no physiological role**. They are xenobiotic persistent organic pollutants (POPs).

Known toxic effects (epidemiological evidence): - Immunotoxicity (reduced vaccine response) - Thyroid disruption - Hepatotoxicity (elevated liver enzymes) - Dyslipidaemia (elevated cholesterol) - Developmental toxicity - Potential carcinogenicity (kidney, testicular cancer – IARC 2B for PFOA)

Catabolism and Elimination Pathway

- PFAS are **not metabolised** in humans (C-F bonds are among the strongest in organic chemistry).
- Long-chain PFAS (PFOS, PFOA): half-lives of **3-8 years** in humans. Eliminated very slowly via urine, faeces, and menstrual blood.
- Short-chain PFAS (PFBS, PFHxA): half-lives of **days to weeks**; faster renal excretion.
- Primary elimination route for short-chain: **urine** (renal clearance).
- Long-chain PFAS bind strongly to serum albumin; minimal urinary excretion.

Expression in Humans

Normal Levels

Compartment	Reference Range
Serum PFOS (general population)	1-20 ng/mL (declining since phase-out)
Serum PFOA	0.5-5 ng/mL
Urine PFOS	<LOD to ~0.1 ng/mL (long-chain: very low in urine)
Urine PFBA/PFBS (short-chain)	0.1-5 ng/mL (more readily excreted)
Serum is primary matrix	Urine is used mainly for short-chain PFAS

Factors Influencing Levels **Increased:** - Proximity to contaminated sites (military bases, industrial facilities) - Occupation (firefighters, fluorochemical workers) - Consumption of contaminated water or food - Use of PFAS-containing consumer products

Decreased: - Avoidance of contaminated sources - Time since last exposure (slow clearance) - Cholestyramine treatment (experimental: enhances faecal excretion)

Associated Pathologies

Condition	PFAS Association	Evidence
Elevated cholesterol	PFOA/PFOS associated with dyslipidaemia	Strong epidemiological evidence
Thyroid disease	PFOS linked to thyroid disruption	Moderate evidence
Immune suppression	Reduced vaccine antibody response (children)	Strong (basis for EPA health advisory)
Kidney cancer	PFOA: IARC Group 2B	Limited human evidence
Testicular cancer	PFOA: elevated risk in exposed workers	Limited evidence
Pre-eclampsia	PFAS exposure linked to pregnancy complications	Emerging evidence

Presence in Urine

Should they be normally present? In the current environment, trace PFAS are detectable in most human urine, reflecting ubiquitous exposure.

Normal urinary levels: Short-chain: 0.1-5 ng/mL. Long-chain: mostly below LOD in urine (better measured in serum).

Form in urine: Ionised species (perfluoroalkyl sulfonates/carboxylates) in their anionic forms.

Solubility: PFAS are amphiphilic (surfactant-like); the ionic species are water-soluble.

Detection in Urine

Available Clinical Assays

1. **LC-MS/MS (gold standard):**
   - **Principle:** Online SPE or protein precipitation + reversed-phase HPLC + tandem MS in negative ESI mode.
   - **Detection:** MRM transitions (e.g., PFOS m/z 499->80; PFOA m/z 413->369; PFBS m/z 299->80).
   - **LOD:** ~0.01-0.1 ng/mL per analyte.
   - **Advantages:** Definitive; multi-analyte (20-50+ PFAS simultaneously); gold standard for biomonitoring.
   - **CDC method:** Per- and Polyfluoroalkyl Substances in Urine and Serum by Online SPE-LC-MS/MS.
2. **UPLC-MS/MS (high-throughput):**
   - Faster chromatography; same MS detection.
   - LOD: ~0.01 ng/mL.

Optimal Urine Type for Measurement

- **Spot urine** with creatinine or specific gravity correction.
- Urine is recommended primarily for **short-chain PFAS** (PFBA, PFBS, PFHxA) which have higher urinary concentrations.

- For **long-chain PFAS** (PFOS, PFOA), **serum/plasma** is the preferred matrix.
- **PFAS-free collection materials** essential (avoid PTFE/Teflon-containing labware).
- Store frozen at -20 C.

Actual Gold Standard

Online SPE-LC-MS/MS (CDC/NHANES method). Measures 20+ PFAS simultaneously in serum and urine. LOD: 0.01-0.1 ng/mL.

Interferences in Measurement

Interference	Effect	Mechanism
PTFE labware contamination	False positive	PFAS leaching from Teflon/PTFE
Matrix effects (ion suppression)	Variable	Co-eluting phospholipids and salts
Background PFAS contamination	Elevated blank	Ubiquitous laboratory contamination (solvents, tubing)
Branched isomers	Multiple peaks	Linear vs branched PFOS require careful chromatography

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- PFAS have minimal UV-Vis absorption (no chromophore in most species).
- **PFOA/PFOS absorption:** Very weak at <210 nm. Not useful for direct UV detection in urine.
- NIR: No useful absorption.

Fluorescence Detection

- Most PFAS are **non-fluorescent**.
- **Indirect competitive immunofluorescence:** Anti-PFOS antibodies + fluorescent tracer. Ex/Em: probe-dependent. LOD: ~0.1-1 ng/mL.
- **MIP-based fluorescence quenching:** PFAS adsorption to molecularly imprinted fluorescent particles. LOD: ~1 ng/mL.
- Still research stage; not validated for clinical use.

Raman Detection

- PFAS have characteristic C-F stretch modes:

Peak (cm-1)	Assignment
~730	CF2 symmetric stretch
~1220	CF2 asymmetric stretch
~390	CF2 wagging

- Conventional Raman: LOD »1 ug/mL (far too insensitive).
- SERS: LOD potentially ~1-10 ng/mL on specialised substrates. Research stage.

FTIR Detection

Band (cm-1)	Assignment	Notes
~1150-1250	C-F stretching (strong, broad)	Characteristic of perfluoroalkyl
~1320-1370	CF2 symmetric stretch	–

- ATR-FTIR can detect PFAS only at mg/L levels. Not useful for ng/L biomonitoring levels.

Voltammetry Detection

- PFAS are **electrochemically inert** in the normal potential window (C-F bonds extremely strong).
- **Indirect electrochemical:** PFAS-selective MIP on electrode reduces capacitance/impedance upon binding.
  - EIS-based sensors: LOD ~0.1-10 ng/mL.
- **Boron-doped diamond electrodes (BDD):** Can reduce some PFAS at very negative potentials (~-2.5 V). Research stage.

Other Detection Technologies

1. **Total organic fluorine (TOF/EOF):** Combustion ion chromatography or extractable organic fluorine. Measures total fluorinated compounds without speciation. LOD: ~2-10 ng/mL F.
2. **Particle-induced gamma-ray emission (PIGE):** Total fluorine. LOD: ~50 ng/mL F.
3. **Immunoassay kits (lateral flow):** Emerging rapid tests for PFOS/PFOA in water. LOD: ~1-10 ng/L. Not validated for urine.
4. **Paper spray MS:** Direct ionisation from paper substrate + MS. LOD: ~0.5 ng/mL. Rapid.

Summary Table: Detection Methods Comparison

Method	LOD	Key Parameter	Sample Prep	Strengths
LC-MS/MS	~0.01-0.1 ng/mL	MRM (ESI-)	SPE/dilute	Gold standard, multi-PFAS
ELISA/immunofluorescence	~0.1-1 ng/mL	450 nm or fluorescence	Plate assay	Screening
Total organic fluorine	~2 ng/mL F	CIC or PIGE	Extraction	Total PFAS burden
EIS-MIP sensor	~0.1-10 ng/mL	Impedance	MIP electrode	Reagent-free
SERS	~1-10 ng/mL	730 cm-1	Substrate	Research
FTIR	>1 ug/mL	1150-1250 cm-1	None	Not practical

References

1. CDC - PFAS in Urine and Serum by Online SPE-LC-MS/MS. https://stacks.cdc.gov/view/cdc/103807
2. Springer ABC - LC-MS for 56 PFAS in plasma. https://link.springer.com/article/10.1007/s00216-026-06486-2
3. Springer IJEHR - Direct PFAS analysis in plasma. https://link.springer.com/article/10.1007/s12403-026-00763-2
4. UniPM - PFAS analytical methods in biological matrices. (Di Giorgi et al.)
5. Academia - PFAS in human urine biomonitoring pilot. https://www.academia.edu/77039978/
6. EPA - PFAS National Primary Drinking Water Regulation (2024).
7. IARC - PFOA and PFOS Monographs (2023).